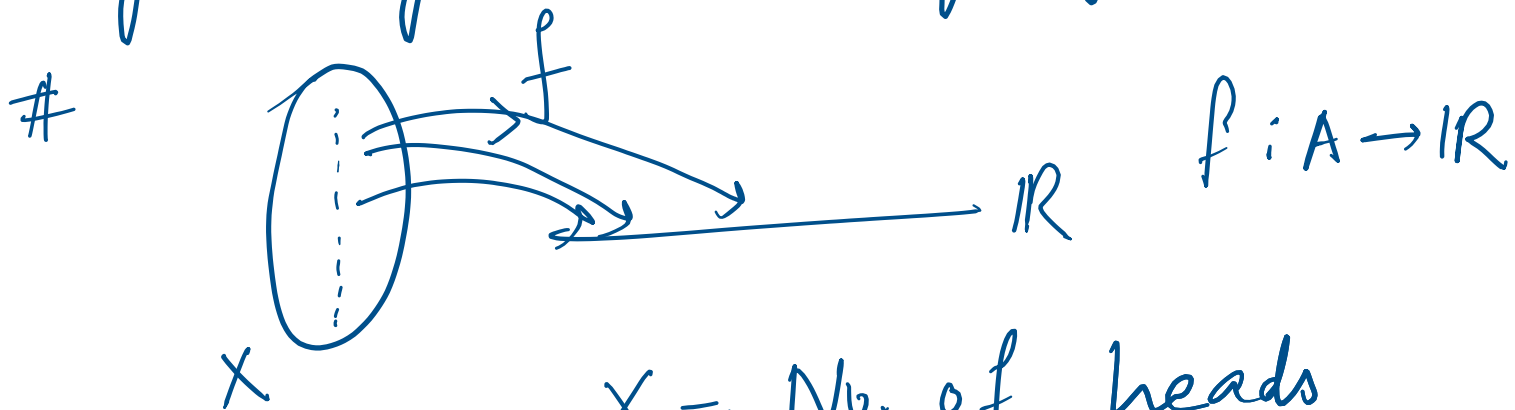


Statistics for Data Science . 27/08/26 .

It is the science of collecting, organising and analysing data.



$X = \text{No. of heads}$
in tossing 2 coins

$$S = \{HH, HT, TH, TT\} \quad X(S) = \{0, 1, 2\}$$

CDF :- $F_X(x) = P(X \leq x)$

(1) $0 \leq F_X(x) \leq 1$

(2) Nondecreasing fⁿ of x

(3) $F_X(-\infty) = 0$, $F_X(\infty) = 1$.

(4) $F_X(x+h) = F_X(x)$ as $h \rightarrow 0^+$

(5) $P(X > x) = 1 - F_X(x) \quad \forall x$

PDF :- $\frac{dF_X(x)}{dx} = f_X(x)$

① $f_X(x) \geq 0 \quad \forall \quad dx$

② $\int_{-\infty}^{\infty} f_X(x) dx = 1$

IID RV :-

↳ independent :-

$$f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2) \dots$$

↳ identical $\Rightarrow f_{X_i}(x_i) = f_X(x) \quad \forall x_i$

Order Statistics :-

→ They are sample values placed in ascending order.

$X_1 = 65 \text{ in}, X_2 = 74 \text{ in}, X_3 = 59 \text{ in}$
 $X_4 = 72 \text{ in}$

$X_3 = 59$
 $X_{(1)}$

$X_1 = 65$
 $X_{(2)}$

$X_4 = 72$
 $X_{(3)}$

$X_2 = 74$
 $X_{(4)}$

Defⁿ:- Let $X_1, X_2, \dots, X_n$ be iid RVs with CDF $F(x)$ and pdf $f(x)$ (assuming all the RVs to be cont). We define

$$X_{(1)} = \min(X_1, X_2, \dots, X_n)$$

$$X_{(2)} = \text{second smallest } (X_1, X_2, \dots, X_n)$$

⋮

$$X_{(n)} = \max(X_1, X_2, \dots, X_n)$$

Then $(X_{(1)}, X_{(2)}, \dots, X_{(n)})$ are called order statistics of $(X_1, X_2, \dots, X_n)$

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

If we are selecting some students based on their scores and we are interested in selecting top 10 students

$$\text{Class score} = (X_1, X_2, \dots, X_n)$$

$$\text{Top 10 students: } - (X_{(n)}, X_{(n-1)}, \dots, X_{(n-9)})$$

$X(k)$ can be different from X_n

$$10, 9, 7, 11 \quad X_2 = 9, \quad X_{(2)} = 9$$

$$(X_1, X_2, X_3, X_4) \quad X_3 = 7, \quad X_{(3)} = 10 \\ \neq (X_{(1)}, X_{(2)}, X_{(3)}, X_{(4)})$$

Order statistics depend only on the order of the values and not on the values themselves.

Recap:- $X = \text{rain in Delhi}$

If you know CDF of RV X , then CDF will be helpful in calculating the prob.

$$P(X < 2 \text{ mm}) = ?$$

$$P(X < 5 \text{ mm}) = ?$$

CDF of each $X_i = F(x)$

PDF of each $X_i = f(x)$

Distⁿ of n th order statistics.

$$X_{(n)} = \max(X_1, X_2, \dots, X_n)$$

$$F_{X_{(n)}}(y_n) = P(X_{(n)} \leq y_n)$$

$$= P(\max(X_1, X_2, \dots, X_n) \leq y_n)$$

indep $\Rightarrow P(X_1 \leq y_n, X_2 \leq y_n, \dots, X_n \leq y_n)$

$\hookrightarrow = P(X_1 \leq y_n) \cdot P(X_2 \leq y_n) \cdot \dots \cdot P(X_n \leq y_n)$

identical $\hookrightarrow = F(y_n) \cdot F(y_n) \cdot \dots \cdot F(y_n)$

$\underbrace{\hspace{10em}}_{n \text{ times}}$

$$= [F(y_n)]^n$$

$$F_{X_{(n)}}(y_n) = P(X_{(n)} \leq y_n) = (F(y_n))^n$$

PDF of nth order statistics:-

$$f_{X_{(n)}}(y_n) = \frac{df_{X_{(n)}}(y_n)}{dy_n}$$

$$\begin{aligned} &= n (F(y_n))^{n-1} \cdot \frac{dF(y_n)}{dy_n} \\ &= n \cdot (F(y_n))^{n-1} \cdot f(y_n) \end{aligned}$$

$\frac{d(g^2(x))}{dx} = 2g(x)g'(x)$

Ques 1 5 RV, $X_1, X_2, X_3, X_4, X_5 \sim U[0, 1]$

CDF, pdf of 5th order statistics?

Soln:- $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{else} \end{cases}$

$$F(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

$$f_{X_{(5)}}(x) = (F(x))^5 = \begin{cases} 0, & x < 0 \\ x^5, & 0 \leq x < 1 \\ 1, & \text{else} \end{cases}$$

$$f_{X(5)}(x) = \begin{cases} 5x^4, & 0 \leq x \leq 1 \\ 0, & \text{else} \end{cases}$$

By direct formula,

$$f_{X(5)}(x) = 5 (F(x))^4 \cdot f(x) \\ = \begin{cases} 5x^4, & 0 \leq x \leq 1 \\ 0, & \text{else} \end{cases}$$

CDF of first order statistics:-

$$X_{(1)} = \min(X_1, X_2, \dots, X_n) \quad \min(2, 3, 4) = 2$$

$$F_{X_{(1)}}(y_1) = P(X_{(1)} \leq y_1)$$

$$= P(\min(X_1, X_2, \dots, X_n) \leq y_1)$$

$$= 1 - P(X_{(1)} > y_1)$$

$$= 1 - P(X_1 > y_1) \cdot P(X_2 > y_1) \cdot \dots$$

$$= 1 - \prod_{i=1}^n P(X_i > y_1)$$

$$\Rightarrow F_{X_{(n)}}(y_1) = 1 - [1 - F(y_1)]^n$$

PDF of $X_{(n)}$:-

$$f_{X_{(n)}}(y_1) = \frac{dF_{X_{(n)}}(y_1)}{dy_1}$$

$$= -n[1 - F(y_1)]^{n-1} \cdot \frac{dF(y_1)}{dy_1}$$

$$= n[1 - F(y_1)]^{n-1} \cdot f(y_1)$$

Q2:- Same RV as Q1. Find the CDF and PDF of $X_{(n)}$.

Solⁿ:- $f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{else} \end{cases}, F(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$

$$\text{CDF of } X_{(n)} = 1 - [1 - F(x)]^n$$

$$= 1 - [1 - x]^5, 0 \leq x \leq 1$$

$$\text{PDF of } X_{(n)} = 5(1-x)^4, 1 = 5(1-x)^4$$

By direct formula,

$$PDF = 5(1-n)^4 \cdot 1 = 5(1-n)^4$$

CDF of $X_{(k)}$ i.e. k -smallest $(X_1, \dots, X_n)$

$$\begin{aligned} F_{X_{(k)}}(y_k) &= P(X_{(k)} \leq y_k) \\ &= P(k\text{th smallest } (X_1, \dots, X_n) \leq y_k) \\ &= P(\text{at least } k \text{ of the } X_i\text{'s} \leq y_k) \\ &= \sum_{j=k}^n P(\underbrace{\text{exactly } j \text{ of } n X_i\text{'s} \leq y_k}_{\text{Binomial}}) \end{aligned}$$

Binomial

$$= \sum_{j=k}^n \binom{n}{j} (F_{X_j}(y_k))^j (1 - F_{X_j}(y_k))^{n-j}$$

$$\begin{aligned} P(A \cup B) &= \\ &P(A) + P(B) - \\ &P(A \cap B) \end{aligned}$$

$$= \sum_{j=k}^n \binom{n}{j} (F(y_k))^j (1 - F(y_k))^{n-j}$$

n, p
 $f(n) = \binom{n}{k} p^k q^{n-k}$

PDF of $X_{(k)}$

$$f_{X_{(k)}}(y_k) = \binom{n}{k} k F(y_k)^{k-1} (1-F(y_k))^{n-k} \cdot f(y_k)$$

$$k=1, f_{X_{(1)}}(y_1) = n (1-F(y_1))^{n-1} f(y_1)$$

$$k=n, f_{X_{(n)}}(y_n) = n F(y_n)^{n-1} f(y_n)$$

Q3:- Same RV as Q1. Find CDF and PDF of $X_{(3)}$?

$$\underline{\text{Soln:-}} F_{X_{(3)}}(u) = \sum_{j=3}^5 \binom{5}{j} u^j (1-u)^{5-j}$$

$$= \binom{5}{3} u^3 (1-u)^2 + \binom{5}{4} u^4 (1-u) + u^5$$

$$= 10 u^3 (1-u)^2 + 5 u^4 (1-u) + u^5$$

$$= 5 u^3 (1-u) (u + 2(1-u)) + u^5$$

$$= 5 u^3 (1-u) (2-u) + u^5$$

$$\text{pdf} = \frac{d F_{X_3}(u)}{du}$$